Descriptive statistics and Table 1

Workflow labs use the variant template: Goal → Approach → Execution → Check → Report.

Learning objectives

- Match the right summary statistic to the right variable type — means for symmetric continuous, medians for skewed, counts and percentages for categorical.
- Build a publication-ready Table 1 with `gtsummary::tbl_summary()`.
- Defend the choice of summary with a quick visual check.

Prerequisites

Labs 1.3 through 1.5.

Background

Table 1 is the first real table of almost every clinical paper. It describes the sample: who was in the study, in how many groups, with what characteristics, and — if the study has arms — how balanced those arms are at baseline. A Table 1 that is built thoughtlessly hides an unbalanced comparison; a Table 1 that is built well anchors the rest of the paper.

The right summary depends on the variable. A continuous variable that is roughly symmetric is summarised by its mean and standard deviation; one that is skewed or has heavy tails is summarised by its median and interquartile range. A categorical variable is summarised by counts and percentages. Reporting the median of a balanced continuous variable is not wrong, but it is unusual; reporting the mean of a skewed variable is often misleading.

gtsummary packages these choices into a declarative API. You say which variables to include and which grouping variable to stratify by, and the table is built with sensible defaults. When you need non-standard behaviour, you override per-variable.

Setup

```r
library(tidyverse)
library(palmerpenguins)
library(gtsummary)
set.seed(42)
theme_set(theme_minimal(base_size = 12))
```

1. Goal

Produce a Table 1 for the `penguins` dataset stratified by species, and back the choice of summary with a diagnostic plot.

2. Approach

```r
  drop_na(species, sex, bill_length_mm, bill_depth_mm,
          flipper_length_mm, body_mass_g)
```

3. Execution

Pick the right summary per variable. First, eyeball the shapes.

```
dat |>
  pivot_longer(c(bill_length_mm, bill_depth_mm,
                 flipper_length_mm, body_mass_g),
               names_to = "variable", values_to = "value") |>
  ggplot(aes(value, fill = species)) +
  geom_histogram(bins = 20, alpha = 0.6, colour = "grey30",
                 position = "identity") +
  facet_wrap(~ variable, scales = "free") +
  labs(x = NULL, y = "Count")
```

All four continuous variables are approximately symmetric within species, so mean (SD) is defensible. If a variable were skewed, we would switch to median (IQR).

```
  select(species, sex, bill_length_mm, bill_depth_mm,
         flipper_length_mm, body_mass_g) |>
  tbl_summary(
    by = species,
    statistic = list(
      all_continuous() ~ "{mean} ({sd})",
      all_categorical() ~ "{n} ({p}%)"
    ),
    digits = all_continuous() ~ 1,
    label = list(
      bill_length_mm    ~ "Bill length (mm)",
      bill_depth_mm     ~ "Bill depth (mm)",
      flipper_length_mm ~ "Flipper length (mm)",
      body_mass_g       ~ "Body mass (g)",
      sex               ~ "Sex"
    )
  ) |>
  add_overall() |>
  add_p()
t1
```

`add_p()` will run a t-test or ANOVA on continuous variables and a chi-square or Fisher's exact test on categorical, with sensible defaults for sample size.

4. Check

Validate the gtsummary output against a hand computation.

```
  group_by(species) |>
  summarise(
    n = n(),
    mean_mass = mean(body_mass_g),
    sd_mass   = sd(body_mass_g),
    .groups = "drop"
  )
```

The group means and standard deviations must match the numbers in the body-mass row of the Table 1.

5. Report

> Table 1 describes `nrow(dat)` penguins from three species in the Palmer Archipelago. Body mass was highest in Gentoo (`round(mean(dat$body_mass_g[dat$species == "Gentoo"]), 0)` ± `round(sd(dat$body_mass_g[dat$species == "Gentoo"]), 0)` g) and lower in Adelie and Chinstrap. All four continuous variables were approximately symmetric within species and are summarised as mean (SD); sex was approximately balanced within each species.

Table 1 is not a tool for hypothesis testing. The *p*-values in the rightmost column describe the data, not the biology. A statistically-significant difference between arms at baseline in a randomised trial is, by construction, a chance finding.

Common pitfalls

- Reporting a mean for a skewed variable.
- Using Table 1 as a place to "prove balance" with *p*-values in a randomised trial.
- Suppressing the missing-data row. Always report it.
- Forgetting that `tbl_summary` drops NAs by default; check the count at the bottom of each column.

Further reading

- Sjoberg DD et al. *Reproducible summary tables with the gtsummary package*. R Journal.
- Altman DG. *Practical Statistics for Medical Research*, chapter on describing data.

Session info

```
```

See also — chapter index

- Methods in this chapter
- Find your method (Chapter 0)